Experimental Study on Stormwater Geyser in Vertical Shaft above Junction Chamber

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Abstract: Stormwater geysers are commonly observed as explosive releases of air-water mixtures from municipal systems, which can cause property damages and human safety concerns. In Edmonton, Alberta, Canada, a number of geyser events were reported in recent years from a 27-m-deep vertical shaft above a junction chamber connecting a 3.5-m-diameter incoming pipe to an outgoing pipe with a 4-m drop. To understand the conditions for the geyser formation and select suitable mitigation methods, an experimental study was conducted using a simplified conceptual 1:20 scaled model. Three series of experiments were conducted with a sudden increase of the inflow rate and different initial flow conditions: Series A with an initial downstream open-channel flow, and Series B and C, both with an initial downstream full-pipe flow, but an air pocket was deliberately entrapped in the upstream pressurized pipe in Series C. A geyser was not observed in Series A mainly due to the available large outflow capacity in the downstream pipe. With the downstream pipe initially full, a geyser event was observed in Series B as pressure surged in the chamber caused by the increased flow rate. Severe geyser events were observed in Series C, with the first phase triggered by the transient pressure wave and the second phase triggered by air released from the air pocket. The relationship between the measured maximum pressure and the geyser height was established. An analytical model is proposed for predicting the magnitude and period of the pressure oscillation induced by the sudden inflow rate increase. The amount of water splashed out of the riser was also measured.
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Introduction

Storm geysers observed as the eruption of air-water mixtures out of manholes can occur in storm sewer systems during intense storm events. They can cause system failures and public safety concerns (Guo and Song 1991; Li and McCorquodale 1999; Zhou et al. 2002). Fig. 1 shows a geyser that occurred in a manhole located at the intersection of 30th Avenue and Gateway Boulevard in Edmonton, Alberta, Canada in July 2016. The height of geyser was almost 17 m, and the duration of the geyser was at least 1 min based on the recorded video. It was the third geyser event at this location since the sewer system was reconstructed in 2012. This manhole has a junction chamber (8.00 m in height and 5.80 m in diameter) under a 27-m-deep vertical shaft (manhole), which connects the eastbound 3.50-m-diameter inflow tunnel and the southbound 3.00-m-diameter outflow tunnel with a 4.00-m drop. In addition, a flow-capacity bottleneck has also been identified in the downstream sewer system, which can cause the aforementioned outflow tunnel to run full. Detailed study is needed to investigate the conditions that could potentially trigger a geyser in this unique manhole with a junction chamber and to test potential geyser mitigation methods.

Storm geysers can form in a vertical manhole shaft (denoted as riser in this study) during the pressurization process of a sewer tunnel from free-surface flow to full-pipe flow when the inflow rate increases. Guo and Song (1990, 1991) presented an analytical investigation on this pressurization process and reported that the intensity of this peak surge (pressure rise) strongly depends on the inflow rate at the time when the tunnel became pressurized. Geysers can also be triggered by the release of the air pocket entrapped in storm drainage systems caused by a rapidly filling bore (Hamam and McCorquodale 1982; Li and McCorquodale 1999; Vasconcelos and Wright 2006). When an air pocket is entrapped in the middle of a horizontal pipe, it may migrate along the tunnel crown in both directions. When it arrives at the partially water-filled vertical shaft, the air release through the shaft can induce a geyser, which involves complex interactions between air and water (Vasconcelos and Wright 2011; Lewis and Wright 2012; Cong et al. 2017). A number of studies also examined the pressure transient during the release of the air pocket driven by a rapidly filling bore (Zhou et al. 2002; Lewis 2011; Zhou et al. 2011; Li and Zhu 2018).

Two kinds of experiments have been conducted in previous studies to investigate geyser formation with a large air pocket entrapped in a horizontal pipe escaping through a partially water-filled vertical shaft. The first one was conducted in a pressurized horizontal pipe with an air pocket suddenly released into the pipe (Vasconcelos and Wright 2011; Lewis and Wright 2012; Cong et al. 2017). When the air pocket migrates and arrives at the water-filled vertical shaft, it enters the shaft, rises, and pushes the water initially in the shaft upward. Meanwhile, the pressure at the shaft bottom gradually drops.

To understand the formation of this type of geyser, the relative upward motion in the riser between the interface of the air pocket and overlying water column and the free surface during the

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Fig. 1. (Color) Geyser event on July 27, 2016, at intersection of 30th Avenue and Gateway Boulevard, Edmonton, Alberta, Canada. (Image courtesy of John Ryalls.)

geysering have been intensively investigated (Vasconcelos and Wright 2011; Lewis and Wright 2012; Cong et al. 2017; Chan et al. 2018). When the air pocket rises steadily at a velocity close to the Taylor bubble velocity, the interface will catch up with the free surface, and the air is released without triggering a geyser (Vasconcelos and Wright 2011; Cong et al. 2017), or the free surface recedes prior to the breakthrough of the air pocket (Muller et al. 2017). The pressure inside the air pocket is mainly determined by the hydrostatic pressure of the overlying water column (Cong et al. 2017). If the air pocket in the shaft rises much faster, it could push the water column in the shaft upward and form a geyser (Chan et al. 2018). During a geyser event, the net rising velocity of the air pocket related to the free surface is much larger than the Taylor bubble velocity, and the pressure inside the air pocket is significantly higher than the hydrostatic pressure caused by its overlying water column (Chan et al. 2018). In addition, it is not essential for the pressure head at the shaft base to exceed the shaft height in order to generate a geyser (Wright et al. 2011a, b; Muller et al. 2017).

Another kind of experiment considered a geyser induced by an entrapped air pocket in a flowing pipe system (Muller et al. 2017; Huang 2017; Huang et al. 2018). Series of experiments were conducted by Huang (2017) to investigate the geyser strength and the pressure variations during geysering under a sudden increase of the inflow rate. Based on the experimental observations, Huang et al. (2018) presented three types of geyser behaviors and discussed the effect of flow-rate increases and riser diameter on geyser heights and pressure peaks.

Given the complexity of air–water interactions in the generation of storm geysers and the unique storm sewer and chamber configurations in this study, a physical model study was conducted to investigate the mechanisms of geyser formation and potential mitigation methods. Experiments were conducted with a sudden increase of the inflow rate under combinations of initial flow conditions. Due to space limitations, the study of possible mitigations has been reported in another paper (Qian et al. forthcoming).

Experimental Setup and Program

The experimental setup was built in the T. Blench Hydraulics Laboratory at the University of Alberta, as shown in Fig. 2. It was a roughly 1:20 scaled model simplified from a prototype sewer system, where the upstream tunnel is connected to the downstream tunnel with a 4.00-m invert drop by a junction chamber, and the top of this chamber is connected to the bottom of the manhole. In the laboratory setup, this chamber was simplified to a $0.30 \times 0.30 \times 0.45$ m (length × width × height) clear PVC cuboid chamber. It was connected at its upstream by a 5.80-m-long acrylic pipe with a diameter of 0.20 m and a slope of 1:100, and its downstream by a 5.95-m-long (L_d) horizontal clear acrylic pipe with a diameter of 0.28 m (D_d). The drop of the upstream and downstream pipe inverts in the chamber is 0.18 m. The clear acrylic vertical riser was connected to the center of the chamber top with a diameter of 0.06 m (d_r) and a length of 1.22 m. A movable flat tailgate was installed at the end of the downstream pipe to control the downstream outflow capacity. A movable circular overflow weir was installed at the downstream tank bottom to control the flow depth in the downstream pipe. Water was fed into the upstream pipe from a pressurized tank.

In the experiments, four pressure transducers (PT) (Model OMEGA MMCG015BIV-10P4C0T3A5CE, OMEGA Engineering, Norwalk, Connecticut) with an accuracy of ±0.2% and range of ±130 kPa were installed to measure both air and water pressures in the system. The positions of the pressure transducers were as follows (Fig. 2): PT1 was installed on the riser wall 0.80 m above the chamber top; PT2 was installed on the chamber top; PT3 was installed on the chamber front wall and 0.02 m above the chamber bottom; and PT4 was installed on the upstream pipe crown and 0.30 m upstream of the chamber. A magnetic flow meter (Foxboro IMT31A, Schneider Electric Systems, Foxborough, Massachusetts) was installed to record the inflow rate. Four video cameras were used during each testing: two cameras (Sony DSC-HX300, Tokyo) for tracking the flow in the upstream pipe at 60 frames per second (fps); a GoPro Hero 5 (San Mateo, California) for recording the movement of the air-water mixture in the chamber and in the riser at 120 fps; and a Sony HDR-PJ58 (Tokyo) for recording geyser height at 60 fps.

To synchronize data collections and video recording, four small bulbs were connected to the ball valve, and at least one bulb was in the view window of each camera. The signals of pressure transducers, flow meter, and bulb currents were simultaneously collected by a data acquisition board (NI-BNC 2120, National Instruments, Austin, Texas) at a frequency of 1,000 Hz. The manual opening of the ball valve would light up four small bulbs of which current was recorded by LabVIEW version 6.0. During each run, the moment when the ball valve was fully opened, the bulb would be lit in the video, which indicates the beginning of the experiment with $t = 0$ s.

To systematically understand the geyser occurrence in the vertical riser above the junction chamber, three series of experiments (total of 36 experiments) were designed based on the possible combinations of the initial steady flow conditions in the upstream and downstream pipes (Table 1). For Series A with downstream open-channel flow, the water depth in the downstream pipe (h_d) was controlled by the overflow weir and initially fixed at one-fourth of the downstream pipe diameter ($h_d/D_d = 1/4$). The initial inflow rate (Q_0) was 20, 30, 40, or 50 L/s, and the final inflow rate (Q_1) was 80, 100, or 120 L/s. For Series B with downstream full-pipe flow, the initial flow in the downstream pipe was also controlled by the overflow weir with $h_d/D_d = 1$, but the chamber was not running full. Q_0 was set as 20, 30, 40, or 50 L/s, and Q_1 was 60, 80, or 100 L/s. For both Series A and B, the initial flow in the upstream

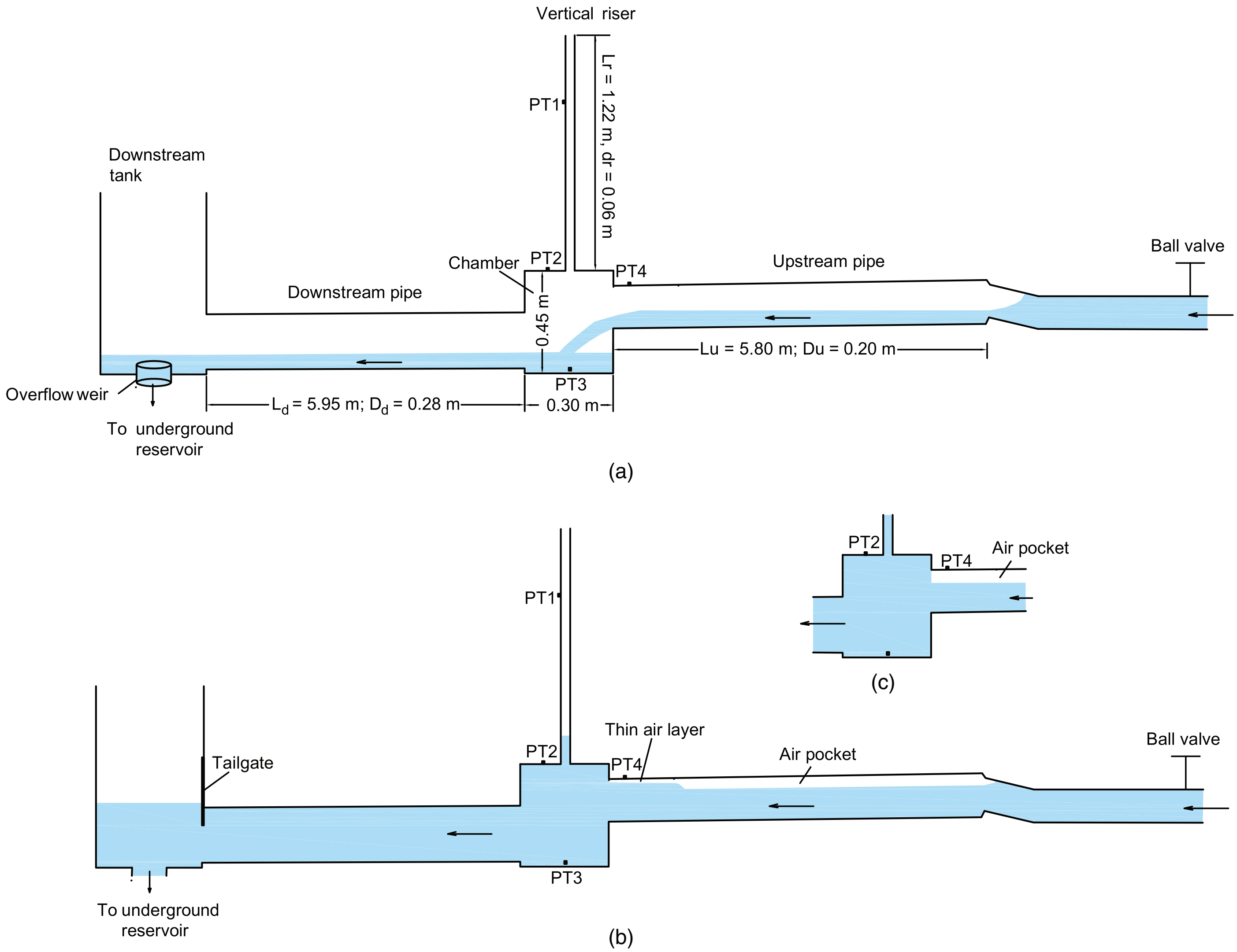

Fig. 2. (Color) Schematic diagram of experimental apparatus: (a) Series A with an upstream free-surface flow; (b) Series C with a partially closed tailgate and an air pocket entrapped in the upstream pipe; and (c) partial zoom of Series C with an air pocket directly connecting to chamber. PT1 to PT4 are the locations of the pressure transducers. Not to scale.

pipe was free-surface flow when $Q_0 = 20$ or 30 L/s, and pressurized pipe flow when $Q_0 = 40$ or 50 L/s. For Series C, the downstream pipe was pressurized by adjusting the tailgate, and the initial water level in the riser (h_{r0}) was preset at 0.1, 0.2, or 0.3 m. An air pocket was also entrapped in the upstream pipe. Initially Q_0 was 15, 20, 25, or 30 L/s, then the inflow rate was suddenly increased to $Q_1 = 40$ L/s. To entrap the air pocket, the tailgate was initially opened to set up the steady free-surface flow for the required Q_0. Then, the tailgate was partially closed, and a surge wave propagated upstream with the downstream pipe and the chamber running full. The tailgate was further adjusted to set up the required water level in the riser, which also pushed the surge wave into the upstream pipe. Thus, the air initially occupying the upper part of the upstream pipe was entrapped as an air pocket.

All experiments were conducted at room temperature of about 20°C. The Reynold's number based on the initial flow in the upstream pipe ranged from 9.5×10^4 to 3.2×10^5 with an initial inflow rate Q_0 from 15 to 50 L/s. The ball valve was manually and suddenly opened for each run, of which the time for fully opening of the ball valve ranged from 0.2 to 0.4 s, which posed a more challenge condition than in the field. Each experiment was repeated three times to ensure the consistency of the results. The experimental results show that the pressure patterns of the three runs were similar, and the discrepancy of the averaged maximum pressures at PT2 was within ±10%. It indicates that the experiments were repeatable.

To simplify the discussion, a geyser in this paper is defined as the release of a water slug or air-water mixture out of the riser top. Jetting height (h) is defined as the maximum height that the air-water mixture can reach during each testing, measured from the riser bottom. If a geyser occurs, the geyser height is the jetting height minus the riser length.

Experimental Results and Discussion

Downstream Open-Channel Flow (Series A)

Fig. 2(a) shows the schematic diagram of the initial flow condition in this series with a free-surface flow in the upstream pipe when $Q_0 = 20$ or 30 L/s. For Case A2, the water depth in the upstream pipe was about 0.08 m. When the inflow rate was suddenly increased from 20 to 100 L/s, a rapid pipe-filling bore formed in

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Table 1. Summary of initial flow conditions of experimental series with a sudden increase of inflow rate

Series	Q_0 (L/s)	Q_1 (L/s)	Initial flow: Downstream pipe	Initial flow: Upstream pipe	h_{r0} (m)
A1	20	80	Open-channel flow	Open-channel	—
A2	20	100	with $h_d = D_d/4$	flow	—
A3	20	120	(controlled by		—
A4	30	80	overflow weir)		—
A5	30	100			—
A6	30	120			—
A7	40	80		Full-pipe flow	—
A8	40	100			—
A9	40	120			—
A10	50	80			—
A11	50	100			—
A12	50	120			—
B1	20	60	Full-pipe flow	Open-channel	—
B2	20	80	(controlled by	flow	—
B3	20	100	overflow weir)		—
B4	30	60			—
B5	30	80			—
B6	30	100			—
B7	40	60		Full-pipe flow	—
B8	40	80			—
B9	40	100			—
B10	50	60			—
B11	50	80			—
B12	50	100			—
C1	15	40	Full-pipe flow	Full-pipe flow	0.1
C2	15		(controlled by	with entrapped	0.2
C3	15		tailgate)	air pocket	0.3
C4	20				0.1
C5	20				0.2
C6	20				0.3
C7	25				0.1
C8	25				0.2
C9	25				0.3
C10	30				0.1
C11	30				0.2
C12	30				0.3

Note: Q_0 = initial inflow rate; Q_1 = final inflow rate; h_d = initial water depth in the downstream pipe; D_d = diameter of downstream pipe; and h_{r0} = height of initial water column in the riser.

the upstream pipe, which propagated toward the chamber at a velocity about 4.56 m/s. It replaced the space in the upper portion of the upstream pipe initially occupied by the air and pressurized the flow behind it. The bore reached the chamber at $t = 1.20$ s and struck the opposite chamber wall at around 1.25 s. The upper part of the bore curved up and back to mix strongly with the remaining air in the upper portion of the chamber, while the lower part directly flowed into the downstream pipe. Strong mixing and swirling were observed in the chamber from 1.25 to 7.00 s, together with the discharge of the air-water mixture from the chamber to the downstream pipe. After 7.00 s, the flow gradually reached a new steady state. A geyser did not happen in this case.

The pressure variations of Case A2 are shown in Fig. 3. At the initial steady state, the pressures in the riser (PT1) and at the chamber top (PT2) were around zero because these locations were exposed to the open air. Only the pressure near the chamber bottom (PT3) had a pressure of 0.99 kPa, which indicated a water depth of 0.10 m in the chamber before the experiment. After sudden increase of the inflow rate, the pressure at PT1 fluctuated around zero because most of the air was pushed into the downstream pipe and expelled through the downstream pipe. The air flow through the riser caused a large oscillation of pressure at PT1 from 2.00 to 6.00 s. The pressure at PT3 began to rise after the filling bore entered the chamber and resulted in an increase of the water depth, and the pressure at PT2 first fluctuated intensively around zero from 1.20 to 2.40 s then gradually rose with fluctuations until 4.00 s. This happened because the air-water mixture in the upper portion of the chamber was first highly entrained with air, then its density increased as the incoming water flow continuously flowed into the chamber.

From 4.00 to 7.00 s, the flow was approaching steady state. The pressures at PT2 and PT3 oscillated at a period around 0.30 s, and their time-averaged value was 2.22 and 4.94 kPa, respectively. The pressure head difference between PT3 and PT2 (0.27 m) was much smaller than their elevation difference (0.43 m). During the final steady state from 7.00 to 14.00 s, the time-averaged pressure at PT2 and PT3 was 2.15 and 4.99 kPa, respectively. The pressure head difference between PT2 and PT3 was also much smaller than their elevation difference due to the effect of dynamic pressure.

For the cases with a larger Q_0 of 40 or 50 L/s, the upstream pipe was initially running full, and only a small part of the upper portion of the chamber was occupied by air. Upon the sudden increase of

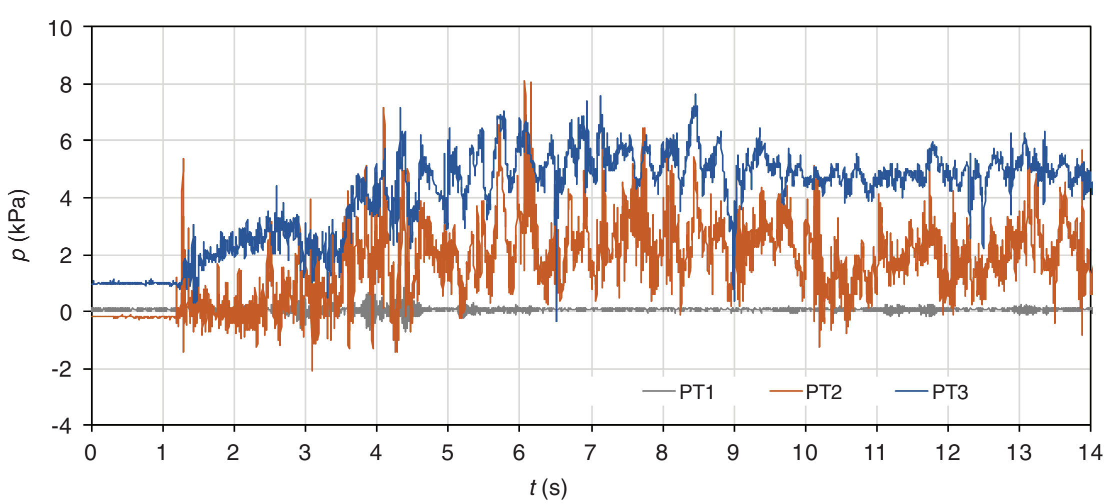

Fig. 3. (Color) Typical case of Series A with downstream open-channel flow: pressure variations at PT1 to PT3 of Case A2 with $Q_0 = 20$ L/s and $Q_1 = 100$ L/s.

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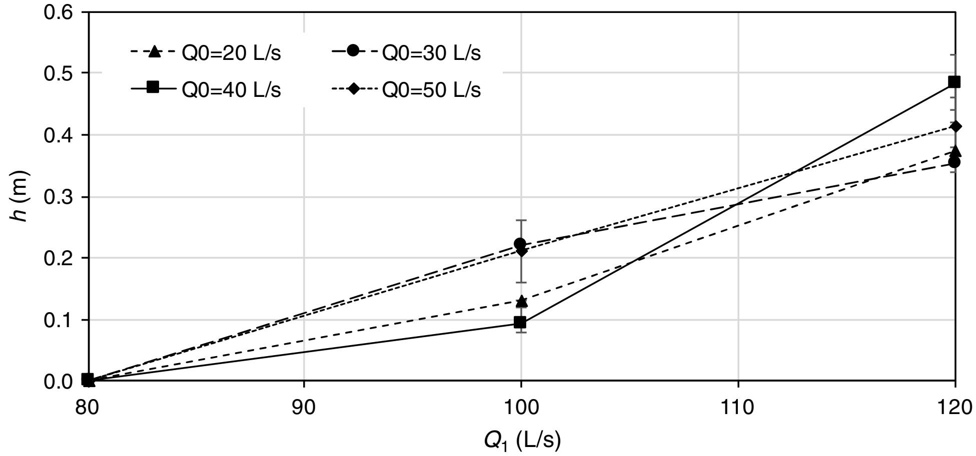

Fig. 4. Heights of the first air-water mixture column appeared in the riser of all cases in Series A.

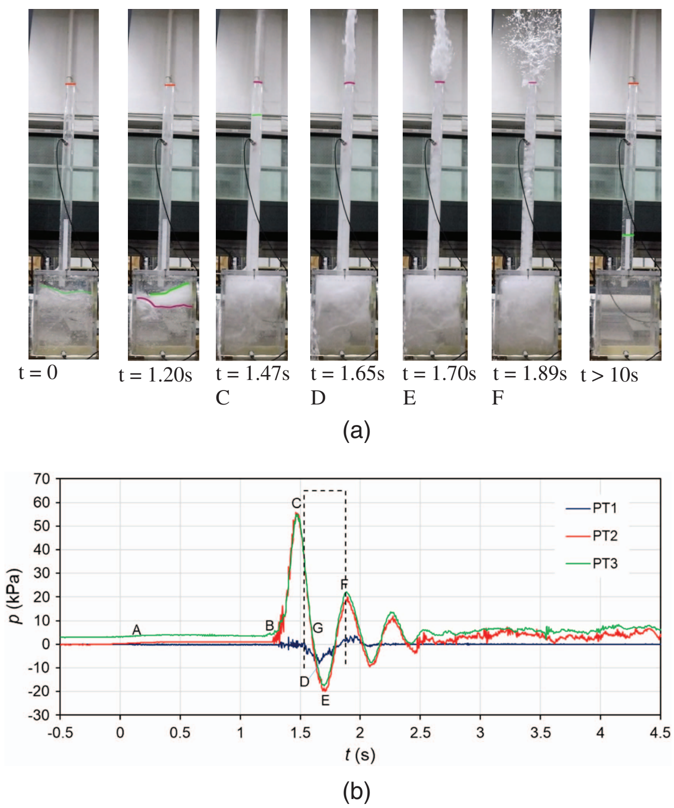

Fig. 5. (Color) Typical case with downstream full-pipe flow (Series B): (a) air-water movement in the chamber and riser (green lines present the free surface of water or air-water mixture in the chamber or riser); and (b) pressure variations of Case B3 with $Q_0 = 20$ L/s and $Q_1 = 100$ L/s. Letters C through F in plot (b) correspond to the moment designated in plot (a).

the inflow rate, the flow from upstream immediately fully filled the chamber, then flowed into the downstream pipe without inducing a large pressure rise.

In this series, a geyser was not observed because the available outflow capacity in the downstream pipe was large enough to pass the increased flow. Both the air initially in the pipes and chamber, as well as the increased flow rate, were easily discharged to the downstream pipe. Fig. 4 shows the measured height h in this series (where h is defined as the maximum height the air-water mixture column first reached in the riser). All heights were lower than the riser height and varied within 0.6 m for $Q_1 = 120$ and 100 L/s. Neglecting the energy loss at the entrance of the riser, the height h then can be roughly estimated using $h = V_u^2/2g - \Delta z$, where V_u is the velocity in the upstream pipe at Q_1, and Δz is the elevation difference between the upstream pipe axis and the chamber top, which is 0.15 m in this study. It gives $h = 0.55$ m for $Q_1 = 120$ L/s and $h = 0.33$ m for $Q_1 = 100$ L/s. These values are about 0.10 or 0.15 m larger than the measured data. From the equation, $Q_1 = 76$ L/s when $h = 0$ m, which is close to the tested value of 80 L/s before h becomes positive. From Fig. 4, Q_1 is found to be the dominant factor in determining geyser height. When Q_1 is the same, a smaller Q_0 indicates a larger flow increase ($\Delta Q = Q_1 - Q_0$) and thus a higher rise.

Downstream Full-Pipe Flow (Series B)

For Series B, the initial downstream flow was full whereas the initial upstream flow could be free-surface flow or full-pipe flow depending on the initial inflow rate Q_0. Fig. 5(a) shows the flow process in the chamber and riser for Case B3, and Fig. 5(b) shows the measured pressure data. The initial conditions of Case B3 differed from that of Case A2 only in the downstream condition. Consequently, the flow before the rapid filling bore reached the chamber ($t = 1.20$ s) was similar except the air initially occupying the upper part of the upstream pipe was expelled through the riser, and a slightly higher pressure was observed at PT2 and PT3 [starting from Point A in Fig. 5(b)]. After 1.20 s, due to the blockage of the downstream full-pipe flow, the flow entering the chamber could not enter the downstream pipe quickly. Thus, it was trapped in the chamber, mixed with the air initially in the chamber, filled and pressurized the chamber. The air-water mixture in the chamber was then pushed into the riser and moved upward. Some mist was observed to first shoot out of the riser at a high speed [$t = 1.47$ s in Fig. 5(a)], then the air-water mixture jetted out the riser top at $t = 1.51$ s as an air-water mixture column similar to the one at $t = 1.65$ s in Fig. 5(a). After a short time, this air-water mixture column began to break and fall downward [$t = 1.70$ to $t = 1.89$ s in Fig. 5(a)], marking the end of the geyser. The free surface in the riser oscillated two more cycles with relatively large magnitude under the influence of inertia, and the new steady state was reached.

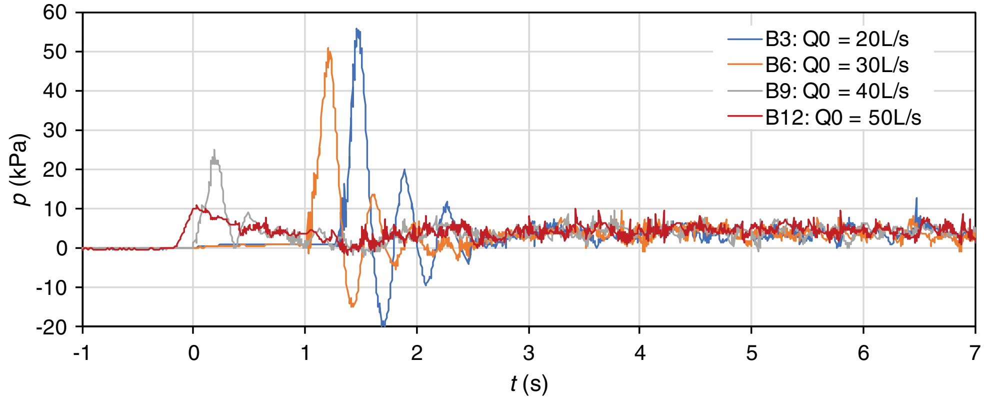

Fig. 6. (Color) Pressure variations at the chamber top (PT2) of cases with final flow rate of $Q_1 = 100$ L/s in Series B.

The pressure variations of Case B3 were different from that of Case A2 after the chamber was completely filled [Point B in Fig. 5(b)]. The pressure at PT1 fluctuated as air or mist was released through the riser, and the pressures at PT2 and PT3 increased rapidly and reached their peak values (55.03–51.76 kPa, Point C) at around $t = 1.47$ s, just before the geyser occurred. During geysering, the pressure at PT1 dropped quickly to its negative peak of −8.30 kPa (Point D), and the pressures at PT2 and PT3 dropped to a negative peak of −20.26 and −17.77 kPa, respectively (Point E). Once the pressure at PT2 was smaller than that of PT1, the air-water mixture in the riser ceased to move up and began to fall down. Due to the continuously incoming flow from the upstream, the pressures at PT2 and PT3 started to rise again. Two more inertia pressure oscillations followed with decreased magnitude and shortened period of 0.38 and 0.34 s. The time-averaged pressures at PT1 to PT3 during the final steady state were 0, 1.82, and 4.65 kPa, respectively, which were close to the time-averaged pressures during the final steady flow for Case A2. However, the pressure at PT3 fluctuated more frequently with a period about 0.17 s and a magnitude around 0.60 kPa. The pressure head difference between PT3 and PT2 (0.29 m) was smaller than their elevation difference (0.43 m), likely due to the dynamic pressure.

Fig. 6 shows the pressure variations at PT2 for all cases with $Q_1 = 100$ L/s in Series B and Q_0 varying from 20 to 50 L/s (Cases B3, B6, B9, and B12). The initial water depth in the chamber increased as the initial inflow rate increased. The pressures at PT2 for both Cases B3 and B6 increased rapidly after the chamber was fully filled, and a geyser was observed. The peak pressure at PT2 for Case B3 (5.57 kPa) was larger than that for Case B6 (5.07 kPa) due to its larger inflow rate increment, but it appeared later because the initial air volume in the chamber was larger. The pressure at PT2 for both cases had one large pressure oscillation followed by two more oscillations with decreasing magnitude and a shortened period. The periods for the first and the second pressure oscillations for two cases were similar, around 0.51 and 0.37 s, but the periods for the third pressure oscillation for two cases were different, 0.37 s for Case B3 and 0.30 s for Case B6. For Cases B9 and B12 with a larger Q_0 of 40 and 50 L/s, both the upstream and downstream pipes were initially full and lightly pressurized. With the sudden increase of the inflow rate, a weak transient pressure wave was induced in the pressurized upstream pipe and traveled faster than the velocities of the filling bores for Cases B3 and B6, which caused immediate pressure rise at PT2. The pressure peaks of Cases B9 and B12 (2.48 and 1.07 kPa, respectively) were much smaller than those of B3 and B6 due to their smaller flowrate increment. Case B9 had two pressure oscillations but Case B12 only had one. At the final steady state, the pressures at PT2 for all four cases oscillated with similar magnitudes and periods because Q_1 was the same.

In this series, the geyser was a single-shoot release of air-water mixture out of the riser. Fig. 7(a) shows that jetting height h correlates linearly with the peak pressures at PT2 (P_{Max}) by $h = 0.6943P_{\mathrm{Max}}/\rho g + 0.3086$ (with R^2 value of 0.97). From the figure, the peak pressure head at PT2 should be 1.31 m upon the geyser occurrence ($h = 1.22$ m). Fig. 7(b) shows the critical ΔQ for triggering a geyser is between 30 to 40 L/s with $h = 1.31$ m. Generally, a geyser occurred when the initial flow in the upstream pipe was a free-surface flow (Cases B1–B3, B5, and B6) or a pressurized pipe flow with ΔQ larger than 40 L/s (Cases B9 and B12). Therefore, it can be concluded that the flowrate increase ΔQ is a key factor leading to a geyser for this series.

Downstream Full-Pipe Flow with Air Pocket Entrapped Upstream (Series C)

To understand the air-water interaction during the geysering in Series C, Case C9 was chosen for the detailed analysis. Fig. 8 shows a few snapshots of the flow process in the chamber and the riser, and Fig. 9 shows the pressure variations. From Fig. 9, the flow process can be divided into two phases. During the first phase, the pressure increased from its initial steady value (Point A) to its final steady value (Point G) in order to pass the increased flowrate, during which two geysers (first and second geysers) occurred and caused pressure fluctuations. During the second phase, the pressure almost stayed the same, but modulated by the six geysers (third to eighth geysers) triggered by the arrival of the air pocket at the chamber.

The initial steady-state flow condition for Case C9 is similar to the schematic diagram shown in Fig. 2(b) with an initial water column of $h_{r0} = 0.30$ m in the riser. An air pocket was entrapped in the upstream pipe. A thin air layer existed along the upstream pipe crown downstream of the air pocket, which connected the air pocket to the chamber and provided a direct passage for the air bubbles to be released from the air pocket into the chamber, which was the source of discrete air bubbles observed in the riser [Fig. 8(a)]. At this initial state, the pressure at PT1 was around zero, and the pressure at PT2 and PT3 was 2.97 and 7.09 kPa, respectively. Their pressure head difference agreed well with their elevation difference. Therefore, the pressure at the riser bottom (PT2) is mainly used to analyze the flow process in the following discussion.

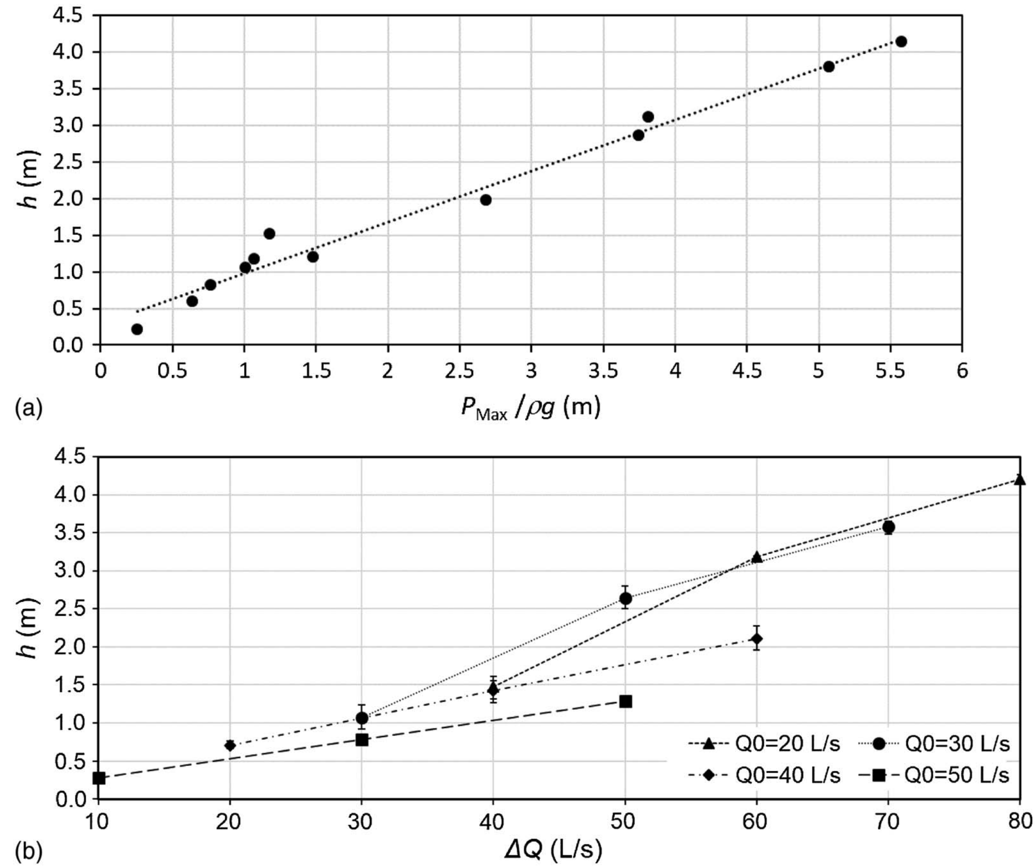

Fig. 7. (a) Relationship between the jetting heights h and peak pressure heads at PT2 ($P_{\text{Max}}/\rho g$); and (b) jetting heights versus inflow rate increases ΔQ for Series B.

The first phase began right after the inflow rate was suddenly increased from 25 to 40 L/s, which trigged a transient pressure. The thin air layer was separated from the air pocket by the interface surge caused by this transient wave, and the air in the thin layer was pushed into the chamber and mixed with the water in the chamber. Meanwhile, the upstream end of the air pocket was pushed downstream slowly by the flow while its downstream end almost stayed around its initial position. The pressure at PT2 immediately began to rise (Point A in Fig. 9) and reached its first peak pressure (10.69 kPa) (Point B) at $t = 0.50$ s, which forced the air-water mixture in the chamber to enter the riser and pushed up the water column initially in the riser. The free surface reached the riser top at $t = 0.73$ s [Fig. 8(b)], and the air-water mixture splashed out to form the first geyser [Fig. 8(c)]. With the release of the air/water mixture, the pressure at PT2 dropped to the minimum (Point D), and the pressure at PT1 also dropped soon after. Air cavity appeared around 1.30 s, became bigger [Fig. 8(c)], and then was directly connected to the air pocket through an air passage along the pipe crown.

Due to the small pressure at PT2, the water in the lower part of the riser began to drop, together with the continuously incoming flow from the upstream pipe, and the chamber was refilled. The air cavity completely disappeared at $t = 1.97$ s, and the downstream end of the air pocket was blocked from the chamber by the water body and continued to stay around its initial position. The pressure at PT2 then quickly rose to its second peak (Point E), and the second geyser occurred. The pressure at PT2 first rose then dropped quickly to its local minimum (Point F), as did the pressure at PT1. At $t = 3.99$ s (Point G), the second geyser ended.

The pressures at PT2 to PT4 remained relatively constant from Point G to Point I in Fig. 9. The free surface in the riser fluctuated near the riser top [Fig. 8(d)] due to the breakup of the discrete air bubbles at the free surface. The air pocket migrated toward the chamber and arrived at the chamber at $t = 6.46$ s (Point I) [schematic plot in Fig. 2(c)], and the pressure inside the air pocket was the same as the pressure at PT4. The second phase started around Point I and included six individual geysers, among which the third geyser was triggered by the large amount of air released from the air pocket due to the arrival of the air pocket, followed by fourth to eighth geysers.

Upon the geyser's occurrence, the pressure head at the riser bottom became smaller than the riser height. After the main body of air was directly connected to the chamber (Point I in Fig. 9), large amount of air was released from the air pocket to the chamber then to the riser due to the larger cross-section area that air occupied [Fig. 2(c)] and a smaller resistance. It induced the rise of the free surface in the riser. The air-water mixture in the riser finally ejected out from the riser top at $t = 6.70$ s to form the third geyser [Fig. 8(e)]. At this moment, the pressure head at PT2 was 0.90 m and lower than the riser height of 1.22 m. It was consistent with the field monitoring result by Wright et al. (2011a, b) and the experimental finding of Muller et al. (2017) that geysering events can occur when the pressure head at the pipe invert is below the grade elevation due to a smaller mixture density.

During the third geyser, the air pressure inside air slug A1 [Fig. 8(f)] was much larger than the hydrostatic pressure caused by the overlying air-water mixture column. Air slug A1 observed in the riser was coalesced from the air bubbles entered the riser

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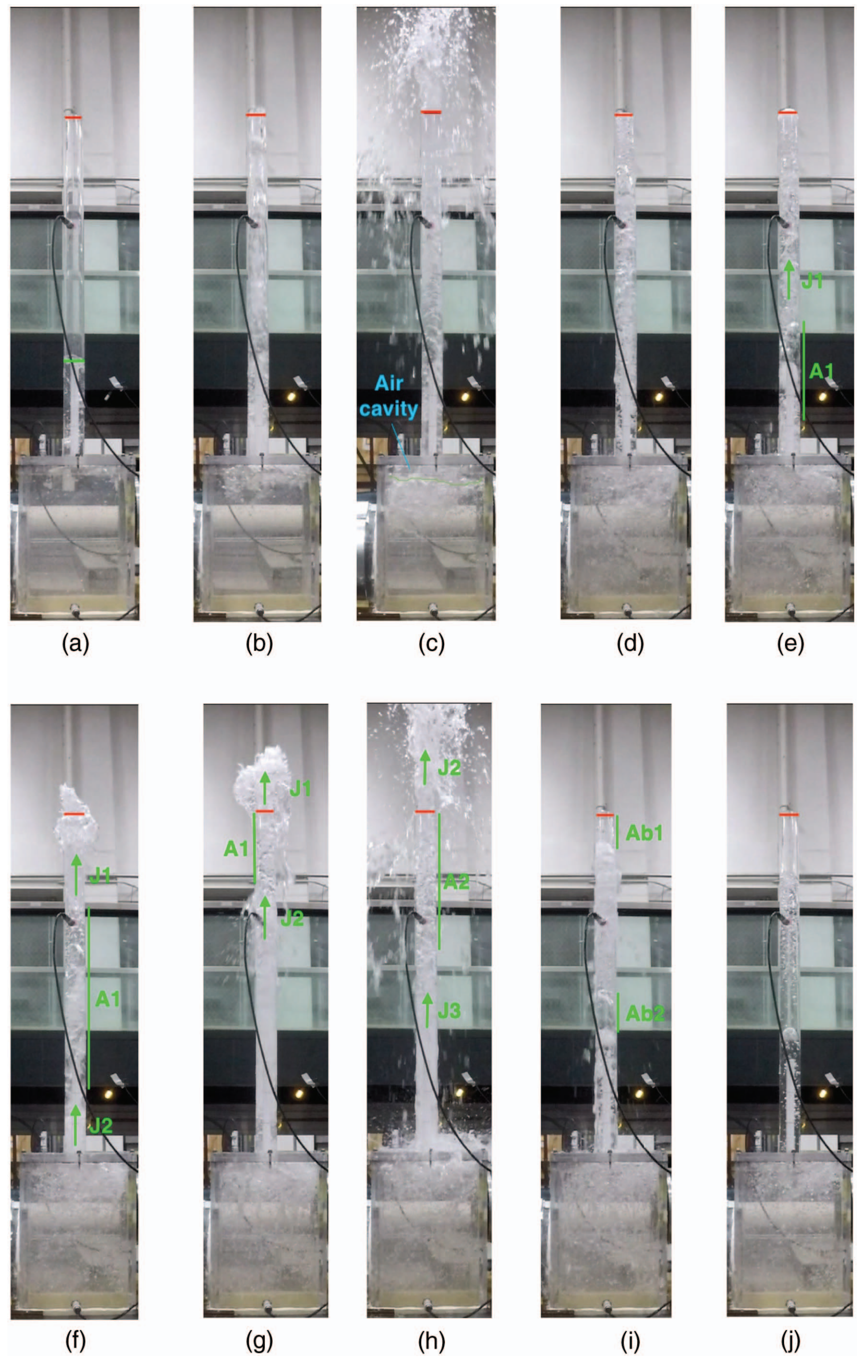

Fig. 8. (Color) Flow process in the chamber and riser for Case C9 with $Q_0 = 25$ L/s, $Q_1 = 40$ L/s, and $h_r = 0.3$ m. A1 to A2 represent air slugs; Ab1 to Ab2 represent large air bubbles; and J1 to J3 represent the front of the recognizable air-water mixture in the riser. Vertical lines along the riser wall denote the range of the large air bubbles or air slugs: (a) $t = 0$ s; (b) $t = 0.73$ s; (c) $t = 1.46$ s; (d) $t = 5.49$ s; (e) $t = 6.70$ s; (f) $t = 7.00$ s; (g) $t = 7.17$ s; (h) $t = 7.50$ s; (i) $t = 14.42$ s; and (j) $t = 19.56$ s.

from the chamber. The interfaces were vague between air slug A1 and the air-water mixture columns both above and under it [Figs. 8(e and f)]. With the geysering, the air-water mixture (above J1) ejected out of the riser, as shown in Fig. 8(f), part of the force acting on the air slug A1 was suddenly released, which resulted in a sudden volume expansion of air slug A1 [comparing air slug A1 in Figs. 8(e and f)]. At this moment ($t = 7.00$ s), the top edge of air slug A1 has just passed PT1, so that the measured pressure head of 0.57 m was the pressure head inside air slug A1. Meanwhile, the air-water mixture height above PT1 was 0.45 m based on the video image. It indicates that the pressure inside air slug A1 could be much larger than the hydrostatic pressure caused by the overlying air-water mixture column during the geysering event.

The aforementioned expansion of air slug A1 in the riser could strengthen the geyser. From $t = 6.70$ s [Fig. 8(d)] to $t = 7.00$ s [Fig. 8(f)], the volume of air slug A1 increased from about 0.91 to 1.70 L. The air pressure inside it dropped by about 46% under the ideal gas law pV = constant for this volume change, where p is the pressure inside the air slug, and V is the volume of the air slug. Consequently, the pressure gradient between the air-water mixture J2 and air slug A1 was greatly increased, which in turn accelerated the air-water mixture J2 going through air slug A1. After air slug

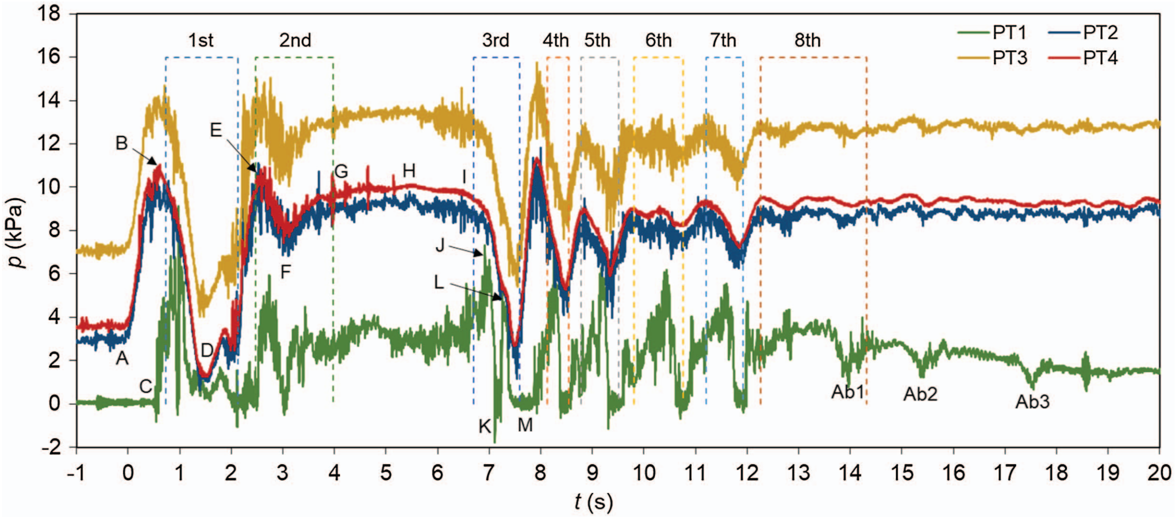

Fig. 9. (Color) Pressure variations for Case C9 with $Q_0 = 25$ L/s, $Q_1 = 40$ L/s, and $h_r = 0.3$ m. Numbering of 1–8 and their frames represent the first to eighth geysers and their duration, respectively; Ab1 to Ab3 identify three large air bubbles, of which Ab1 and Ab2 are represented in Fig. 8(d).

A1 arrived at the riser top, it broke and was completely exposed to the atmosphere [Fig. 8(g)], the pressure inside air slug A1, i.e., pressure at PT1, dropped to its minimum (Point K in Fig. 9). The air-water mixture J2 was accelerated again to a velocity of 5.75 m/s when it was jetting out of the riser. The mixture reached a much higher height compared with the jetting height of the air-water mixture J1 [Fig. 8(g)]. In this study, air slug A1 in the riser expanded during the geysering and accelerated the air-water mixture below it, whereas the air pocket in the riser studied by Chan et al. (2018) was compressed and accelerated the water column overlying it during the geysering.

The third geyser ended at $t = 7.59$ s. Considering the elevation difference between PT2 and PT4 (0.05 m), air could only be released from the air pocket to the chamber due to buoyancy because the piezometric head difference between PT4 and PT2 was almost zero. With the continuously incoming water from the upstream, the air pocket was compressed, the pressure inside the air pocket (PT4) increased as shown in Fig. 9. As a result, large amount of air was again released to the chamber. The free surface in the riser began to rise again and jetted out of the riser at $t = 8.12$ s as the fourth geyser. At this moment, the pressure at PT4 and PT2 was 9.33 and 6.93 kPa respectively. The approximate piezometric head difference between PT4 and PT2 was 0.19 m, which could induce large amount air release from the air pocket to support the formation of the fourth geyser. A new cycle began, followed by the fifth to eighth geysers.

Different from the earlier studies, during the third to eighth geysers, the main body of the air pocket remained in the upstream pipe and released air bubbles into the chamber. Some air bubbles in the chamber then entered the riser and coalesced as air slugs in the riser. The vague interface between the air slug and the overlying air-water mixture column in the riser never caught up with the free surface in the riser. The air slug always broke at the riser top during the geysering. This indicates that the geyser formation in this study could not be analyzed by the relative upward motion of the interface and the free surface in the riser, but this multigeyser event is similar to the one recorded in the field in Minneapolis, Minnesota (Wright et al. 2011a, b).

After the last geyser, the free surface in the riser was lower than the riser top, while the air bubbles in the riser continued to migrate upward and broke at the free surface. The complex bubbly flow in the riser was approaching to a single-water-phase condition with discrete air bubbles entrained [Fig. 8(j)]. The flow reached its final steady state after about 19.00 s. The average final steady pressure at PT2 to PT4 was 8.79, 12.76, and 9.25 kPa, respectively, with the difference due to the hydrostatic pressure.

During the late stage of Series C, large air bubbles with a distorted spherical cap were observed in the riser [e.g., Ab1 and Ab2 in Fig. 8(i)] when the flow gradually approached to steady state. Three well-defined air bubbles for Case C9 were selected from the video to compare with the Taylor bubble velocity ($= 0.345\sqrt{gd_r}$, where g is gravitational acceleration) (Davis and Taylor 1950). The net rising velocities of these three air bubbles were 0.61, 0.55, and 0.35 m/s, relative to the free surface. All were larger than the Taylor bubble velocity of 0.26 m/s. All bubbles reached the free surface in the riser then broke without triggering geyser, but the free surface lowered. It indicates that even though the net rising velocity of the large air bubble is larger than the Taylor bubble velocity, a geyser does not always occur. This result differs from that of Cong et al. (2017), in which an alternative definition of a geyser was proposed as the net air pocket velocity is greater than the Taylor bubble velocity. When these air bubbles migrated upward in a zigzag path, the height of the overlying air-water mixture column decreased due to the film flow. When the upper edge of these bubbles arrived at PT1, the pressure at PT1 began to drop (Ab1 to Ab3 in Fig. 9) due to the decreasing of the mixture column height above it. When the lower edge arrived at PT1, the pressure at PT1 reached its minimum value, then began to rise as the mixture height above PT1 was increasing with the rising of the air bubble.

Peak pressures at PT2 for all cases in Series C are shown in Fig. 10, in which P_{1m}, P_{2m}, and P_{4m} are the peak pressure of the first, second, and fourth geyser, respectively (e.g., Case C9 in Fig. 9), and P_{Final} is the final steady pressure at PT2. For Cases C1–C3, all of P_{4m} are missing because those cases could not reach the steady flow phase (i.e., second phase) due to a small tailgate opening and P_{Final} being larger than the riser height (1.22 m); thus, water was continuously flowing out of the riser. Fig. 10 also shows that for each individual case, P_{1m} and P_{2m} are close to each other, larger than P_{4m} for Cases C4–C6, and comparable with P_{4m} for Cases C7–C12. It indicates that the maximum pressure at PT2 (P_{Max}) in Series C could occur during the first phase (e.g., for Cases C1–C6), or during the second phase (e.g., for Cases C7–C12).

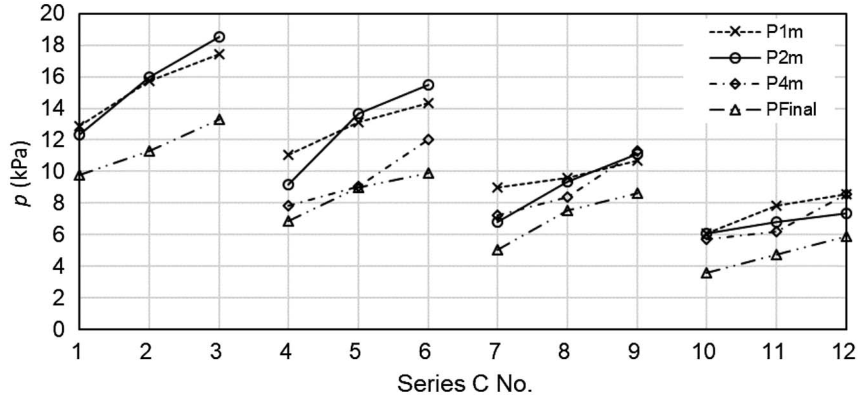

Fig. 10. Variations of the peak pressures (P_{1m}, P_{2m}, and P_{4m}) of the first, second, and fourth geysers, and the final steady pressure at PT2 (P_{Final}) for each case in Series C.

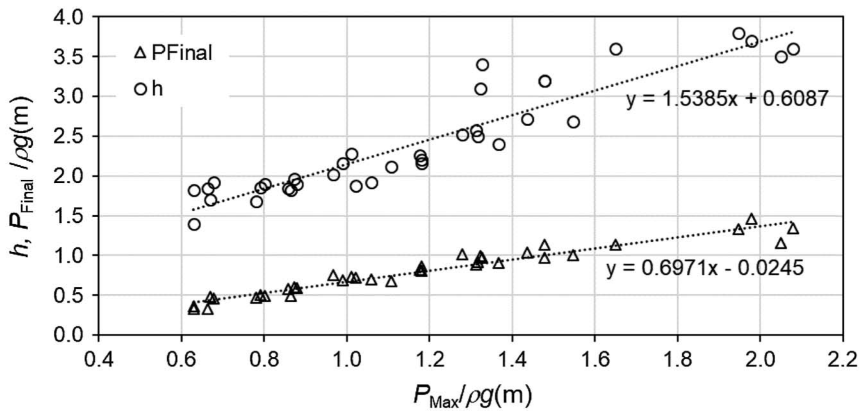

Fig. 11. Series C: final steady pressure heads at PT2 ($P_{Final}/\rho g$) and the jetting heights (h) versus maximum pressure heads at PT2 ($P_{Max}/\rho g$).

All P_{1m}, P_{2m}, P_{4m}, and P_{Final} decrease with the decrease of ΔQ at the same h_{r0}, and increases with the increase of h_{r0} at same Q_0. Both a small Q_0 and a large h_{r0} indicate a small tailgate opening when other flow conditions are the same. A small tailgate opening results in a small outflow capacity, thus a larger peak pressure for the steady flow (P_{Final}). Therefore, when the flow capacity downstream in a drainage system is limited, it will result in a relative large peak pressure and is more likely to induce a severe geyser if an air pocket was entrapped in the system, e.g., Case C3.

The relationships between P_{Max} and P_{Final} as well as the jetting height (h) are shown in Fig. 11. Both h and P_{Final} increase linearly with P_{Max}. The linear correlation coefficient between h and P_{Max} is larger than 1.0, which indicates that the air-water mixture out of the riser is highly aerated. It also shows that P_{Final} is only about $(0.56–0.76)P_{Max}$. The positive correlation between P_{Final} and P_{Max} indicates that P_{Max} also decreases with the decrease of ΔQ. Therefore, for Series C, both h and P_{Max} can be characterized by P_{Final}, i.e., the outflow capacity downstream of the manhole.

Spilled Water Volume

In order to gain a basic understanding of how much water splashed out from the riser, rough measurements were made for some typical cases. The results are summarized in Table 2, in which V_1, V_2, and V_3 are the spilled water volumes for three repeated runs, and V_{avg} is the averaged value. For all runs, the differences of water amount between the individual run and the average are within $\pm 15\%$, which means that experiments were repeatable.

For cases in Series B, the splashed water amount mainly depends on Q_0 and ΔQ. For Cases B3, B6, and B9 with $Q_1 = 100$ L/s and $\Delta Q = 80$, 70, and 60 L/s, the spilled water volume V_{avg} is 0.72, 0.58, and 0.29 L, respectively. The V_{avg} increases with the increasing of ΔQ but is relatively small (less than 1.00 L) compared with its flowrate increase ΔQ in Series B. The initial flow in the upstream pipe is free-surface flow for Cases B3 and B6, but pressurized pipe flow for Case B9.

For all cases in Series C with the same $Q_1 (= 40$ L/s), both a smaller Q_0 or a higher h_{r0} can result in more water splashing out

Table 2. Spilled water volumes for typical cases in Series B and C

Case No.	Q_0 (L/s)	Q_1 (L/s)	h_r (m)	V_1 (L)	V_2 (L)	V_3 (L)	V_{avg} (L)
B3	20	100	—	0.65	0.78	0.82	0.72
B6	30	100	—	0.63	0.65	0.55	0.58
B9	40	100	—	0.22	0.22	0.27	0.29
C4	20	40	0.1	10.8	9.30	8.02	9.70
C7	25	40	0.1	1.55	1.65	1.80	1.60
C8	25	40	0.2	4.10	3.80	4.42	4.45

Note: V_1, V_2, and V_3 = spilled water volume of first, second, and third repetitions, respectively; and V_{avg} = averaged spilled water volume of three repetitions of each case.

due to the higher P_{Final} caused by a smaller Q_0 or a higher h_{r0}, as discussed previously. For Cases C4 and C7 with the same $h_{r0}(= 0.1\ \mathrm{m})$, V_{avg} increases by about five times from 1.60 to 9.70 L when ΔQ increases from 15 to 20 L/s. For Cases C7 and C8 with the same Q_0 and ΔQ, V_{avg} is 4.45 L for Case C8, nearly three times that of Case C7, when h_{r0} increased from 0.1 to 0.2 m. V_{avg} in Series C is much larger than that in Series B (Table 2). For example, V_{avg} for Case C4 is over 12 times greater than that of Case B3 when they have the same $Q_0 = 20$ L/s but different initial upstream flow conditions, because multiple geysers happened and lasted over 10.00 s for Case C4 whereas only a single geyser occurred and lasted less than 1.00 s for Case B3.

Characteristics of Pressure Oscillations

To understand the pressure oscillations caused by the sudden increase of the inflow rate, a simplified analytical model is proposed in this section. To simplify the analysis, the following assumptions are made: (1) the upstream and downstream pipes and the chamber are initially filled with water; (2) the water flow is incompressible; (3) the energy loss including the frictional loss and entrance/exit loss are negligible, so is the velocity head; and (4) the riser is long enough without spilling. Then, the control volumes in the downstream pipe and in the riser can be selected as CV1 and CV2, respectively, as shown in Fig. 12. Thus, the momentum equations for control volumes CV1 and CV2 can be written

$$\frac{L_d}{gA_d}\frac{dQ_d}{dt} = H_{ch} - H_E \tag{1}$$

$$\frac{h_r}{gA_r}\frac{dQ_r}{dt} = H_{ch} - H_s \tag{2}$$

and the continuity equations are

$$Q_u = Q_d + Q_r \tag{3}$$

$$Q_r = A_r\frac{dH_s}{dt} \tag{4}$$

where t = time; A_d and A_r = cross-sectional area of the downstream pipe and riser, respectively; Q_u, Q_d, and Q_r = flowrate in the upstream pipe, downstream pipe, and riser, respectively; H_{ch}, H_E, and H_s = piezometric head in the chamber center, at the outlet center of the downstream pipe, and at the free surface in the riser, respectively; and h_r = water column height in the riser. H_{ch} in Eqs. (1) and (2) refers to slightly different locations but is treated the same with the assumption of negligible energy losses.

As shown in Eq. (1), a higher pressure head in the chamber (H_{ch}) is required in order to accelerate the flow in the downstream pipe to pass more flow. The excess flow ($Q_r = Q_u - Q_d$) will cause the increase of the pressure in the chamber and the increase of water level in the riser. To obtain the pressure variation, subtract Eq. (2) from Eq. (1) to cancel H_{ch} and apply Eqs. (3) and (4)

$$\left(\frac{h_r}{gA_r} + \frac{L_d}{gA_d}\right)A_r\frac{d^2H_s}{dt^2} + H_s - H_E - \frac{L_d}{gA_d}\frac{dQ_u}{dt} = 0 \tag{5}$$

Eq. (5) cannot be solved analytically due to its nonlinearity because h_r in the coefficient in the first term is related to H_s, as shown in Fig. 12. Therefore, Eq. (5) is linearized by assuming h_r in the coefficient as the initial water column height h_{r0} at $t = 0$. Assuming dQ_u/dt is a constant and equal to $(Q_1 - Q_0)/T_v$ (where T_v is the duration of the inflow increase from Q_0 to Q_1), and the piezometric head at the downstream end H_E is a constant as H_{E0}, then the general solution to Eq. (5) is

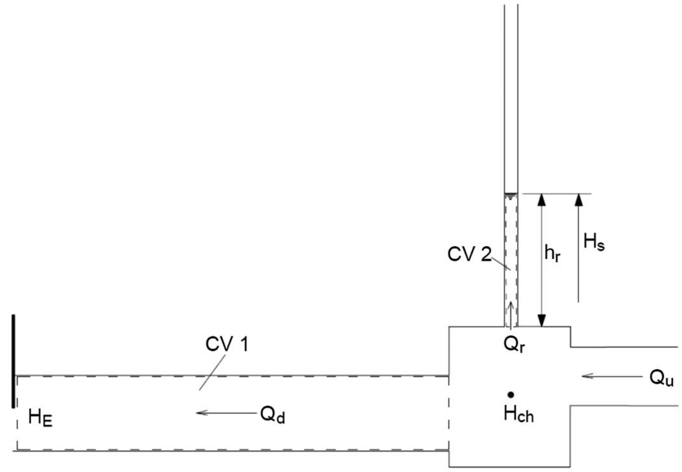

Fig. 12. Control volumes CV1 and CV2 in schematically experimental setup for analytical analysis.

$$H_s = H_E + \frac{L_d}{gA_d}\frac{dQ_u}{dt} + C_1\cos\left(\frac{2\pi}{T}t + C_2\right) \tag{6}$$

where C_1 and C_2 = constants. Applying the initial condition at $t = 0$, $H_s = H_{s0} = H_{E0}$ due to the negligible energy loss, and $Q_r = 0$, then H_s can be expressed

$$H_s = H_{s0} + \frac{L_d}{gA_d}\frac{dQ_u}{dt}\left[1 - \cos\frac{2\pi}{T}t\right] \tag{7}$$

where the period of the free oscillation of the water column in the riser T is

$$T = 2\pi\sqrt{\frac{L_dA_r}{gA_d} + \frac{h_{r0}}{g}} \tag{8}$$

The first term in Eq. (8) (L_dA_r/gA_d) is similar to that derived for a conventional surge tank, except that the downstream pipe length (L_d) and cross section (A_d) are used here, while the pipe characteristics upstream of the surge tank are used in the surge tank equation. This is because in this study, the flow rate in the upstream pipe is given, whereas the flow rate in the downstream pipe can vary. For a traditional surge tank, the transient flow occurs between the upstream reservoir and the surge tank without downstream outflow. The second term in Eq. (8) (h_{r0}/g) reflects the inertial and gravitational effect of water column initially in the riser. Its effect on T is up to the relative value between h_{r0}/g and L_dA_r/gA_d, e.g., a large h_{r0} and a small L_d, means a large effect on T. Guo and Song (1991) derived a similar equation to Eq. (8) for a dropshaft-drift tube system without outflow from the system. They had a third term ($4L_d^2/\pi^2a^2$) considering the effect of the flow compressibility, where a is the speed of pressure wave. In this study, the flow in the downstream acrylic pipe is almost pure water flow; so that, the speed of the pressure wave in the downstream pipe is about 305 m/s (Wylie et al. 1993). This third term is only about 0.6% of the first term; thus, the effect of compressibility is negligible in this study.

Fig. 13 compares the measured piezometric head at PT2 and the predicted H_s by Eq. (7) after adjusting the datum to be the chamber top for three cases in Series C with a sudden increase of the inflow rate. The duration of the flow rate increase T_v is set as 0.4 s based on the duration of the valve opening. In general, both the period and peak of the pressure predicted by Eqs. (7) and (8) agree quite well

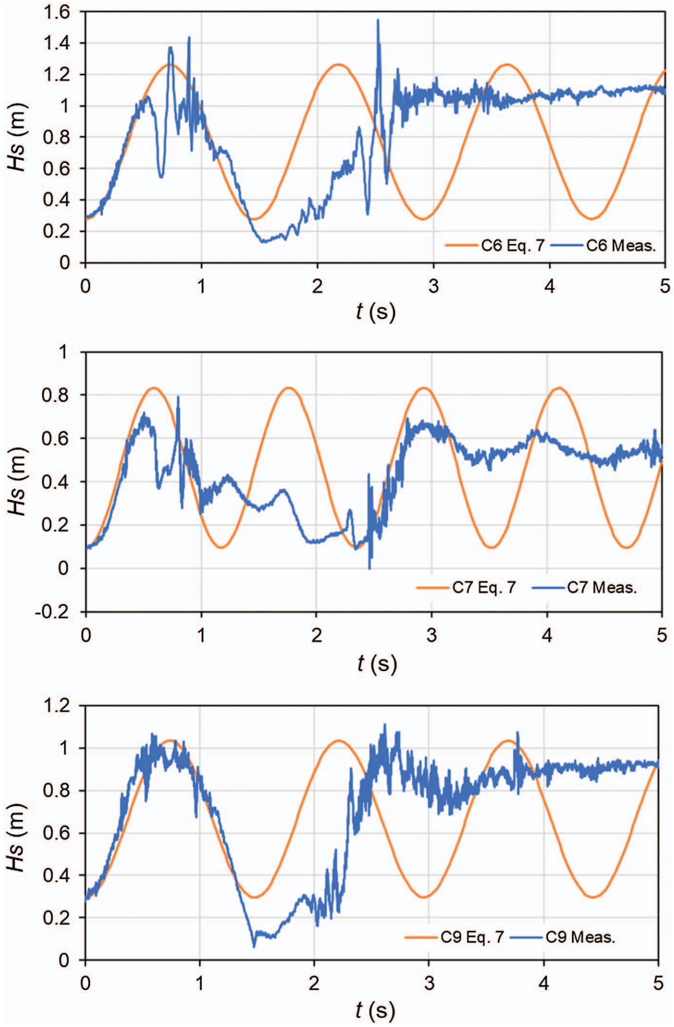

Fig. 13. (Color) Measured and predicted [by Eq. (7)] variations of piezometric heads at PT2 for Series C caused by the sudden increase of the inflow rate. The datum is set at the chamber top.

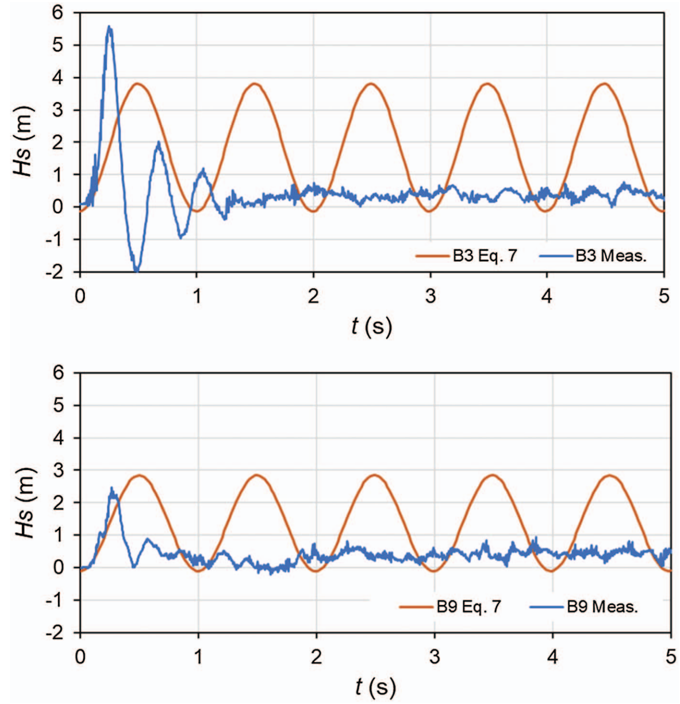

Fig. 14. (Color) Measured and predicted [by Eq. (7)] variations of piezometric heads at PT2 for Series B caused by the sudden increase of the inflow rate. The datum is set at the chamber top. The time $t = 0$ is set as the time when the pressure begins to rise.

with the measurements of the first geyser for Series C. For Cases C3, C7, and C9, the predicted period T by Eq. (8) is 1.45, 1.17, and 1.45 s, respectively. The periods for Cases C3 and C9 are the same because they have same $h_{r0} = 0.30$ m, and higher than that for Case C7, where $h_{r0} = 0.10\,\mathrm{m}$. For the experimental setup in this study, the surge tank solution for all cases is the same with $T = 0.99$ s, much smaller than the measurements. Thus when h_{r0} is large, its effect on the period of pressure oscillation can be significant.

When the pressure at PT2 drops near its minimum, it begins to deviate from the predicted variation of H_s. A number of reasons can contribute to this difference. For example, the solution is a linearized equation from its initial flow condition without energy dissipation. Additionally, Eq. (7) was derived assuming water in the riser and chamber is an incompressible single phase without air, whereas an air cavity appears in the chamber in the experiment around the time when Pressure at PT2 is near its minimum [Fig. 8(c)]. The appearance of the air cavity extends the duration of the minimum pressure range until the space occupied by the air cavity is completely refilled by water.

The predictions of the piezometric pressure head at PT2 by Eq. (7) for Series B are compared with the measurements in Fig. 14 after adjusting the datum to be the chamber top and with $T_v = 0.40$ s. The chamber for both Cases B3 and B9 was initially not full, and thus h_{r0} in Eq. (7) was set to be zero. The figure shows that both the magnitude and period vary significantly from the measurements. The predicted period (0.99 s) is about twice that of the measured one for Case B3 (0.52 s) and three times of Case B9 (0.37 s), and the predicted pressure magnitude (3.80 m) for Case B3 is about two-thirds of the measured one (5.57 m). The main reason for these discrepancies is that the upstream pipe and chamber for Case B3 were initially not full, and thus it does not satisfy the assumption for Eq. (7) of a pressurized upstream pipe. A large negative pressure appears in the measurement for Case B3 whereas negative pressure does not occur in the predicted one. It is mostly because the rapid filling bore in the unfull upstream pipe has an air-water mixture front. After this air-entrained water fully fills the chamber, the fluid in the chamber becomes compressible, which in turn induces the negative pressure in the experiments. For Cases B9, the negative peak does not appear for both the measured and predicted data due to its pressurized upstream pipe.

As the pressure in the chamber is increased in order to accelerate the flow in the downstream pipe, Eq. (1) can be integrated to find out the required momentum for the increase of downstream discharge

$$\int (p - p_E) A_d dt = \rho L_d A_d \frac{\Delta Q}{A_d} \tag{9}$$

where p = pressure at the inlet center of the downstream pipe, and is assumed equal to the pressure in the chamber; p_E = pressure at the outlet center of the downstream pipe; ρ = density of water; and ΔQ = flowrate increase in the downstream pipe (as the flow rate in the riser is typically much smaller, as discussed previously).

Eq. (9) indicates that the net resultant impulse acting on the water body in the downstream pipe equals its momentum change.

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Assume p_E is a constant with an initial value of $p_{E0} = p_0$ (where p_0 is the initial pressure in the chamber), the net resultant impulse can then be roughly estimated by integrating the positive part of the first large pressure oscillation [e.g., from Point B to Point G in Fig. 5(b)]. The momentum change in the downstream pipe can be estimated using Eq. (9) with the flow rate increasing from Q_0 to Q_1 (ΔQ). Thus, the net resultant impulse and the momentum change are about 500 and 479 $\mathrm{kg \cdot m/s}$ for Case B3, 484 and 415 $\mathrm{kg \cdot m/s}$ for Case B6, and 321 and 356 $\mathrm{kg \cdot m/s}$ for Case B9. The impulse and momentum change for each case are close to each other with a discrepancy within $\pm 13\%$. Thus, in order to pass an increased discharge, a net impulse (through an increased pressure in the chamber) is needed to increase the outflow momentum flux. A longer downstream pipe, a larger downstream pipe diameter, or a larger discharge increase requires a larger net impulse, i.e., a larger pressure in the chamber or a longer time duration.

Summaries and Conclusions

In this study, a large-scale physical model was built to understand the mechanism of geysers occurring in a vertical riser above a junction chamber that connects upstream and downstream pipes with different invert elevations. Three series of experiments were conducted with a sudden increase of the inflow rate: an initial downstream open-channel flow, or an initial downstream full-pipe flow with/without chamber being surcharged. For Series A, a geyser does not occur because the available outflow capacity is large enough for the increased flow rate, and the current riser height is also large. For Series B, a single-shoot geyser can be triggered by the rapidly filling bore if the increased inflow rate is large enough and the initial free-surface flow in the upstream pipe can induce a large negative pressure in the chamber. For Series C, with the limited outflow capacity downstream and an air pocket entrapped upstream, the geyser can be triggered in the riser first by the transient pressure wave (first phase), then by the air released from the air pocket (second phase). The second phase usually includes several individual geysers that involve strong air-water interactions in the riser, and this air-water interaction may strengthen the geyser. During the geysering, the expansion of the air slug in the riser can accelerate the air-water mixture beneath it. At the late stage of the second phase, large air bubbles may appear in the riser, rise at a velocity larger than the Taylor bubble velocity, and break at the free surface in the riser without triggering a geyser.

All geyser heights in Series B and C positively correlate with the peak pressures in the chamber. The jetting heights in Series B are comparable with the peak pressure heads at the riser bottom, and the jetting heights in Series C are much higher than the peak pressure heads at the riser bottom. The downstream outflow capacity in Series C has great effect on the peak pressure and the geyser height. The water amount splashed out the riser during the geysering for the cases in Series B is much smaller than that in Series C.

An analytical model was proposed to estimate the period and the magnitude of the pressure oscillation for the study system caused by the linear increase of the inflow rate assuming that the system is pressurized and the energy loss is negligible. The linearized analytical model predictions of the magnitude and period agree well with the measurements of the first pressure oscillation in Series C. The analytical model is not applicable to the cases in Series B because the upstream pipe and the chamber were not full. The application of the momentum equation for Series B shows that the pressure process inside the chamber is to provide the required resultant impulse force to accelerate the flow in the downstream pipe to pass the increased flow rate. A downstream pipe of a larger diameter or a longer length, or a larger discharge increase will require a larger net impulse, i.e., a larger pressure in the chamber or a longer time duration.

Data Availability Statement

Some or all data, models, or code generated or used during the study are available from the corresponding author by request (experimental measurements).

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Notation

The following symbols are used in this paper:

A_d = cross-sectional area of downstream pipe;
A_r = cross-sectional area of vertical riser;
a = speed of pressure wave;
C_1, C_2 = constant coefficients;
D_d = diameter of downstream pipe;
d_r = diameter of vertical riser;
g = gravitational acceleration;
H_E = piezometric head at outlet center of downstream pipe;
H_{ch} = piezometric head at chamber center;
H_s = piezometric head at the free surface in riser;
h = jetting height;
h_d = water depth in downstream pipe;
h_r = water column height in riser;
L_d = length of downstream pipe;
P_{1m}, P_{2m}, and P_{4m} = peak pressure of the first, second and fourth geyser for Series C, respectively;
P_{Final} = final steady pressure at PT2;
P_{Max} = maximum peak pressure at PT2;
p = pressure;
p_E = pressure at outlet center of downstream pipe;
Q_d = flowrate in downstream pipe;
Q_r = flowrate in riser;
Q_u = flowrate in upstream pipe;
Q_0 = initial inflow rate;
Q_1 = final inflow rate;
T = period of free oscillation of water column in riser;
T_v = increasing time of inflow rate from Q_0 to Q_1;
t = time;
V = volume of air slug;
V_{avg} = averaged spilled water volume of three repeated runs;
V_i = spilled water volume of ith repeated run of each case;
V_u = velocity in upstream pipe;
ΔQ = increase of inflow rate ($= Q_1 - Q_0$);

Δz = elevation difference; and
ρ = density of water.

Subscript

0 = initial value of variables at $t = 0$ s.

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